Predicting Diabetes Risk from Health Indicators Using Machine Learning

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***Abstract* - Diabetes is a rapidly growing public health concern, and early detection is key to preventing severe health complications. Traditional diagnosis methods often rely on clinical laboratory tests, which can be costly and inaccessible for many individuals. This study explores whether self-reported health indicators from the CDC Diabetes Health Indicators dataset, obtained through the UCI Machine Learning Repository, can be used to accurately predict diabetes risk. The dataset consists of 253,680 responses from the 2015 Behavioral Risk Factor Surveillance System (BRFSS), with 21 features covering medical, lifestyle, and demographic attributes. After data preparation, visualization, feature selection, and model training, multiple machine learning algorithms were evaluated. Gradient Boosting achieved the best performance, with an accuracy of approximately 85% and an AUC of 0.8156. The findings show that BMI, high blood pressure, high cholesterol, age category, and difficulty walking are among the strongest predictors. This research demonstrates the potential of large-scale survey data to support early diabetes risk assessment and public-health decision-making.**

Keywords - Diabetes prediction, machine learning, health indicators, Gradient Boosting, Random Forest, feature selection, data visualization..

I. Introduction

Diabetes remains one of the most widespread chronic health conditions in the United States and continues to grow in prevalence each year. More than 38 million individuals are currently living with diabetes and millions more are estimated to be in a prediabetic state without being formally diagnosed. The long-term complications of diabetes affect multiple organ systems including the heart kidneys eyes and nervous system. These complications result in substantial personal and economic burdens. Early detection of diabetes and timely identification of individuals at elevated risk are therefore crucial for reducing preventable complications and improving long term quality of life.

Traditional methods for diagnosing diabetes rely on clinical laboratory measurements such as fasting plasma glucose, oral glucose tolerance tests, and A1C levels. Although these procedures are medically reliable, they require clinical infrastructure, specialized personnel, and financial resources that may not be accessible to all communities. For this reason, researchers and public health professionals have become increasingly interested in the possibility of predicting diabetes risk using self-reported survey data. Such data are widely available, cost effective to collect at scale, and can be used to screen large segments of the population long before clinical symptoms emerge.

This study uses the CDC Diabetes Health Indicators dataset from the UCI Machine Learning Repository which contains 253,680 survey responses collected through the Behavioral Risk Factor Surveillance System in 2015. The dataset includes 21 features covering chronic conditions, lifestyle habits, psychological and physical wellbeing, functional limitations, and demographic descriptors. These variables represent many of the observable factors that contribute to long term metabolic health and risk of diabetes. Because the dataset is large and diverse, it provides a strong foundation for applying machine learning models to understand population level diabetes patterns and to predict individual risk.
A central objective of this research is to formally examine three key research questions.

R1. Which medical and lifestyle indicators—such as BMI, blood pressure, cholesterol levels, and physical activity—are most influential in predicting diabetes?
R2. Can machine learning models accurately predict the presence of diabetes using health indicators from the BRFSS dataset?
R3. What demographic patterns, such as age, education, and income levels, differ between diabetic and non-diabetic individuals?

Initial exploratory analysis suggests that obesity cardiovascular risk factors physical inactivity and poor general health demonstrate strong associations with diabetes. These findings are consistent with long established medical research and suggest that self-reported data contain meaningful patterns that can be extracted through machine learning. By evaluating multiple models including Logistic Regression Decision Tree Random Forest and Gradient Boosting the study aims to determine which modeling approaches provide the strongest predictive performance and the most reliable interpretation of the underlying health factors.

This work contributes to the growing body of research that leverages machine learning for public health analytics. By demonstrating that diabetes risk can be predicted using survey-based indicators, the study highlights the potential for developing scalable early risk detection tools that support preventive healthcare initiatives. The insights gained from this research can guide public health agencies in designing targeted interventions and encouraging screening among individuals who are most vulnerable to developing diabetes.

II. OVERVIEW OF THE ANALYSIS PROCESS

A. DATA AND INFORMATION ACQUISITION

The dataset used in this study was sourced from the UCI Machine Learning Repository and originates from the CDC Behavioural Risk Factor Surveillance System for the year 2015. It contains 253,680 survey entries each representing a single individual. The dataset includes 21 features covering medical lifestyle and demographic attributes and 1 binary label known as Diabetes binary which identifies whether the person has diabetes or prediabetes.

This dataset was selected because it is publicly available large in size and widely used in research. It provides broad coverage across different regions age groups income groups and education levels. Since the dataset consists entirely of self-reported information it serves as a practical resource for building scalable and cost effective early risk detection models.

Summary statistics:

	Diabetes_binary	HighBP	HighChol	CholCheck	BMI	Smoker	Stroke	HeartDiseaseorAttack	PhysActivity
count	253680.000000	253680.000000	253680.000000	253680.000000	253680.000000	253680.000000	253680.000000	253680.000000	253680.000000
mean	0.139333	0.429001	0.424121	0.962670	28.382364	0.443169	0.040571	0.094186	0.756544
std	0.346294	0.494934	0.494210	0.189571	6.608694	0.496761	0.197294	0.292087	0.429169
min	0.000000	0.000000	0.000000	0.000000	12.000000	0.000000	0.000000	0.000000	0.000000
25%	0.000000	0.000000	0.000000	1.000000	24.000000	0.000000	0.000000	0.000000	1.000000
50%	0.000000	0.000000	0.000000	1.000000	27.000000	0.000000	0.000000	0.000000	1.000000
75%	0.000000	1.000000	1.000000	1.000000	31.000000	1.000000	0.000000	0.000000	1.000000
max	1.000000	1.000000	1.000000	1.000000	98.000000	1.000000	1.000000	1.000000	1.000000

Fig. 1. Sample rows from the dataset

B. DATA AND INFORMATION PREPARATION

Before model training several preparation steps were performed. The dataset does not contain missing values so no imputation was required. Since all variables are numeric the data is compatible with standard machine learning models without additional encoding. A major challenge is that the dataset is imbalanced as a majority of individuals do not have diabetes. This imbalance can cause models to favor the majority class and misclassify individuals with diabetes. To correct this class weighting was applied during model training so that errors in predicting diabetic individuals are penalized more heavily.

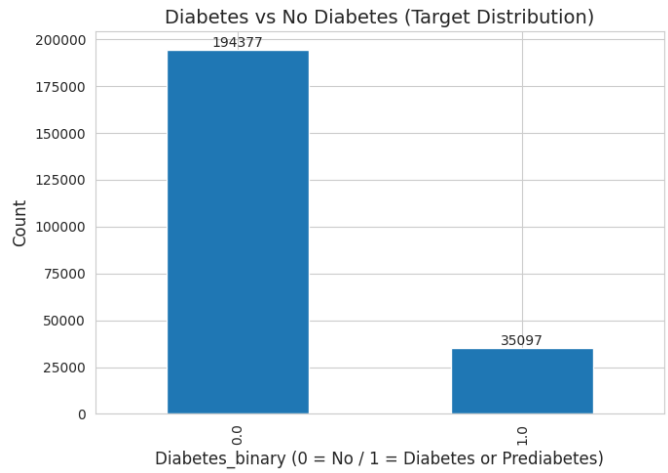

Fig. 2. Target class distribution.

The dataset was then divided into training and testing subsets to evaluate model generalization performance. Exploratory data analysis was conducted to understand the distribution of features and their relationships with the target label.

C. INFORMATION MODELING AND VISUALIZATION

Information modeling and visualization played an essential role in understanding the underlying structure of the dataset and in identifying the relationships between various health indicators and diabetes. The exploratory analysis began with continuous variables, where several clear and meaningful patterns emerged. Individuals classified as diabetic consistently exhibited higher BMI values compared to non diabetic individuals. This trend reflects well documented medical findings that excess body weight is one of the strongest risk factors for diabetes. Visual inspection of the BMI distribution revealed a noticeable shift toward higher ranges for the diabetic group.

Mentally unhealthy days and physically unhealthy days also showed significant differences. Individuals with diabetes reported a higher number of unhealthy days per month, suggesting that diabetes affects not only physical wellbeing but also mental health. These visualizations highlighted the burden of chronic illness and provided early evidence that overall health condition is closely tied to diabetes prevalence. The patterns in these continuous variables helped establish a foundation for the feature selection and modeling stages.

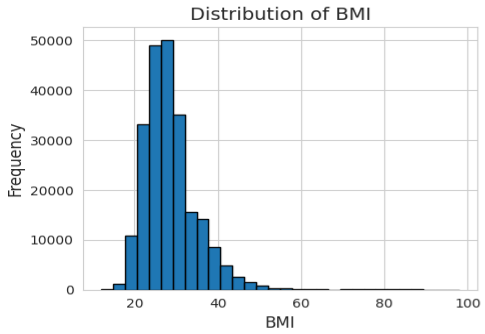

Fig. 3. Distribution of BMI among all respondents.

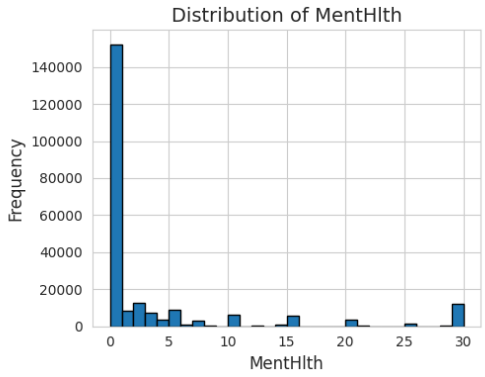

Fig. 4. Distribution of reported mentally unhealthy days.

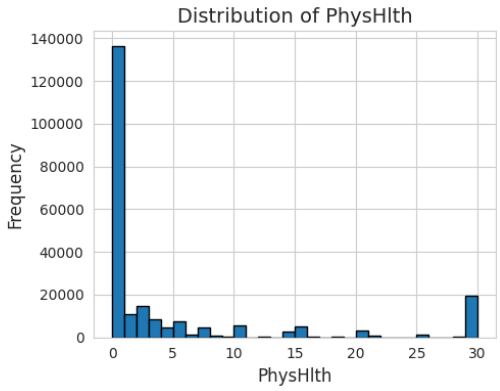

Fig. 5. Distribution of physically unhealthy days.

To further understand the structure of the dataset, several binary medical indicators were visualized. These indicators provided insights into the presence or absence of specific conditions among respondents. The distributions showed that individuals with diabetes were far more likely to report high blood pressure and high cholesterol. Since diabetes is closely related to cardiovascular risk, the elevated frequency of these conditions validated the reliability of the dataset. Smoking status was also examined and although the differences were not as pronounced as in blood pressure or cholesterol, smokers still appeared more frequently within the diabetic group. Stroke history and heart disease were more common among individuals with diabetes, which again reflects clinical observations that diabetes increases cardiovascular vulnerability. Physical activity showed an inverse pattern, where diabetic individuals reported lower participation in regular physical exercise. Together these binary indicators provided a richer understanding of health profiles and demonstrated that diabetes is often associated with multiple coexisting medical conditions.

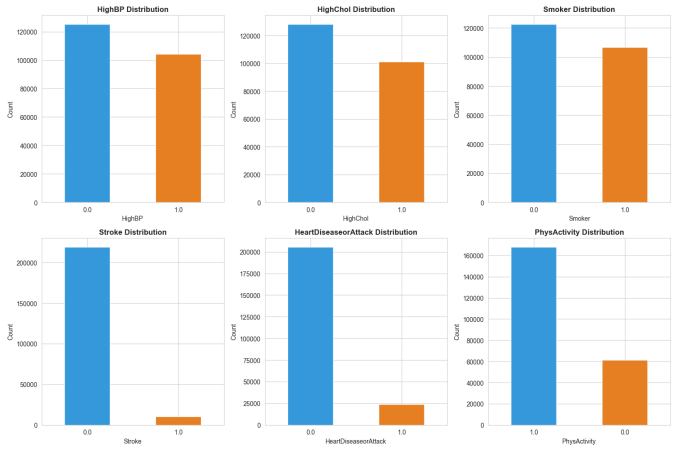

Fig. 6. Key health indicators

Demographic variables were also examined to understand the representation within the dataset. While these distributions do not reflect direct diabetes prevalence, they reveal important characteristics about the dataset composition. The age distribution indicated that the dataset contains a larger number of individuals in older age groups which is valuable because diabetes becomes increasingly common with age. Education levels were broadly distributed with a higher representation of individuals with moderate to high educational attainment. Income distribution increased steadily across the categories suggesting that the dataset captures a diverse range of socioeconomic backgrounds. By understanding these demographic distributions, the study was able to interpret later findings in the context of population representation.

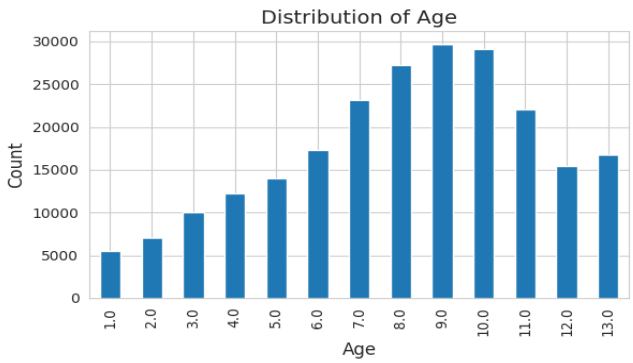

Fig. 7. Distribution of respondents across age groups.

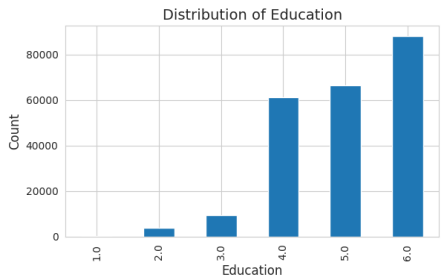

Fig. 8. Distribution of educational attainment levels.

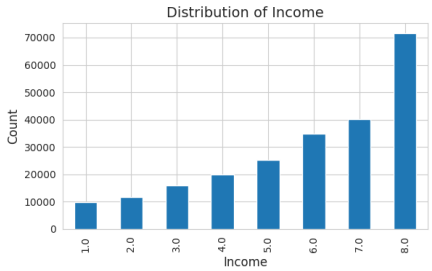

Fig. 9. Income distribution of surveyed respondents.

The correlation analysis provided a summary of how variables relate to each other and to the diabetes label. The correlation heatmap highlighted meaningful associations between diabetes and several important health indicators. BMI high blood pressure high cholesterol general health and difficulty walking showed the strongest correlations with the diabetes variable. Although the correlation values were not extremely high due to the complexity of health data and the binary nature of many variables, the patterns were consistent with established clinical research. The heatmap also showed that some variables were correlated with each other such as high blood pressure and high cholesterol which often occur together. These insights informed the feature selection stage and ensured that the most relevant predictors were chosen for modeling.

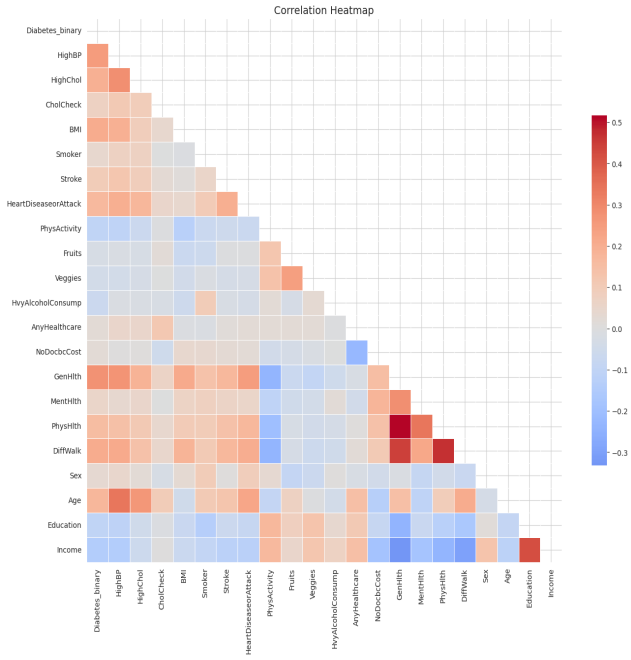

Fig. 10. Correlation heatmap of health and demographic indicators.

Overall the information modeling and visualization stage provided a comprehensive understanding of the dataset. It revealed important health trends consistent with clinical literature and helped guide subsequent stages including feature selection and model development. The exploratory analysis demonstrated that the dataset contains meaningful patterns that machine learning models can learn effectively.

III. FEATURE SELECTION

Feature selection is an essential step in the development of reliable predictive models because it improves model performance reduces overfitting and simplifies interpretation. In this study feature selection was carried out using two statistical approaches known as Chi Square and Mutual Information. Both techniques evaluate the strength of association between individual predictors and the diabetes label although they rely on different mathematical principles. The Chi Square test measures how the distribution of a feature varies across the target classes. Features that show a strong dependence on the diabetes label produce higher Chi Square scores. In this dataset the test ranked variables such as BMI physically unhealthy days mentally unhealthy days difficulty walking and age category among the top predictors. These findings align with clinical understanding because poor physical health higher BMI and mobility challenges are common among individuals with diabetes.

Mutual Information was applied as a complementary method to capture nonlinear relationships between predictors and the

target variable. This method evaluates how much information a feature contributes to reducing uncertainty in predicting diabetes. The results showed that high blood pressure general health cholesterol check physical activity and high cholesterol exhibited substantial predictive value. Interestingly both feature selection techniques converged on a common set of influential variables. This convergence strengthens confidence in the selected features and supports the validity of the dataset. After evaluating the results from both techniques the top 15 features were chosen for model training. This selection ensured that the models were trained using the most meaningful inputs while excluding features that contributed little predictive value or introduced noise.

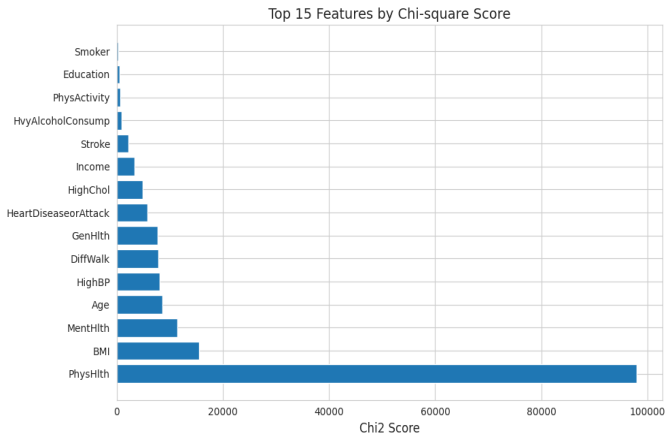

Fig. 11. Top features ranked by Chi-Square scores.

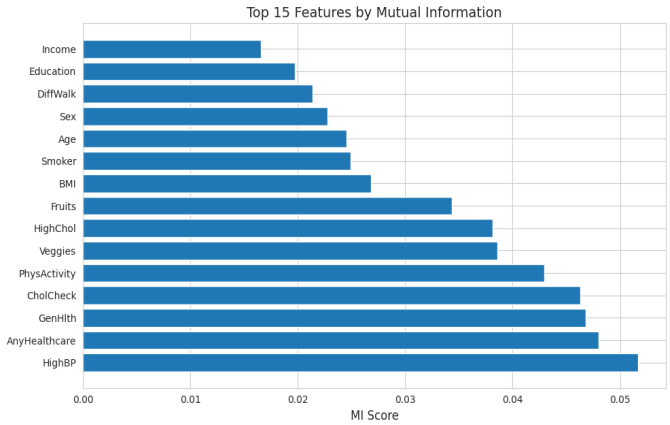

Fig 12. Top features ranked by Mutual Information scores.

IV. MODELING AND EVALUATION

Four machine learning models were developed to evaluate the predictive value of self-reported health indicators. These models represent a progression from simple linear decision boundaries to sophisticated ensemble learning methods. Training each model involved using the selected top 15 features and applying class weighting to address class imbalance. Performance was evaluated using confusion matrices accuracy values and ROC AUC metrics.

A. Logistic Regression

Logistic Regression was used as the baseline model for this research. Logistic Regression also serves as the simplest model where complexity from the other models can be evaluated against. The model predicts that the contributions of the predictors to the log odds of diabetes increase linearly, which leads to a straightforward and clear decision boundary. The model was able to demonstrate satisfactory performance by being able to range individuals in order of diabetes risk. The model even achieved a small AUC of 0.80 which is representative of the model being able to differentiate between positive and negative diabetes cases at many levels. The confusion matrix showed that the model was able to predict non diabetic individuals with relative ease, however the model showed only moderate performance in spotting the diabetic individuals which is normal across datasets where there is a large imbalance of cases. Overall, Logistic Regression was a great model to establish as a baseline for this research.

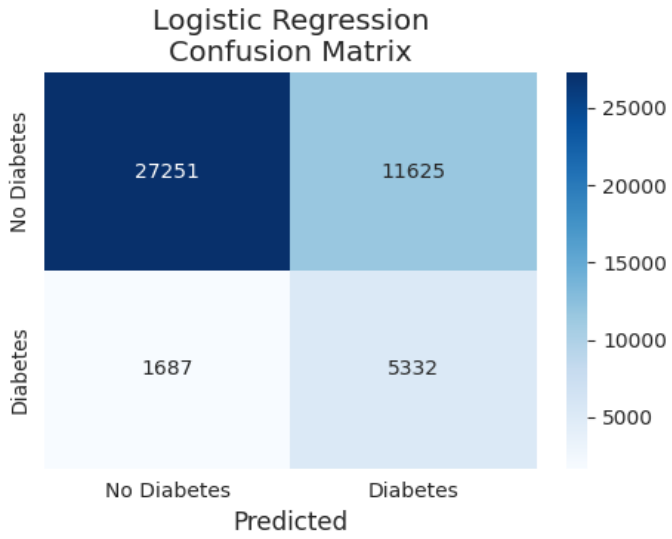

Fig. 13. Confusion matrix of Logistic Regression model.

B. Decision Tree

The Decision Tree Learning model contains nonlinear associations in the form of iterative divisions of the dataset based on thresholds of the characteristics to maximize information gain. Such a setup makes the model highly interpretable since the decision-making process is clearly

stated. The main problem in the modeling technique, however, lies in the overfitting problem common in a wide range of datasets with many categorical features. In the experiment, the generalization capability of the Decision Tree Learning model was relatively poor. Its AUC value was poorer compared to the ensemble techniques. The confusion matrix also pointed to a lack of strong capability for distinguishing subjects who have diabetics. The technique was more adept at modeling noise in the dataset than the overall trend of the populations.

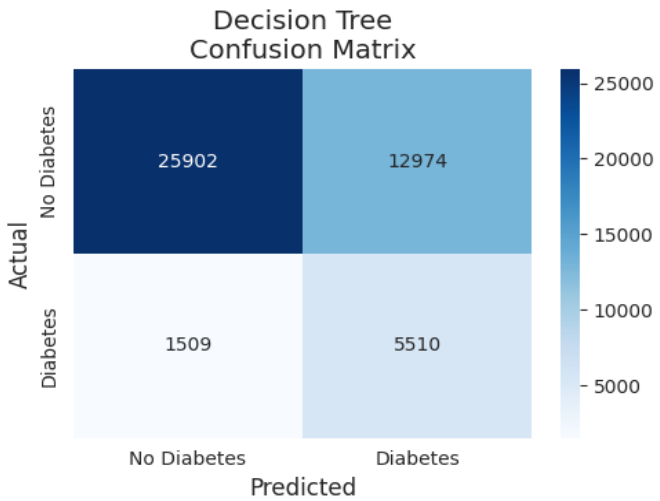

Fig. 14. Confusion matrix of Decision Tree model.

C. RANDOM FOREST

Random Forest is an ensemble approach that combines many Decision Trees predictions, each trained on different slices of the data. By averaging the outputs of numerous trees, it reduces overfitting and captures more stable patterns. Its performance considerably exceeds the performance of a single Decision Tree. This can be shown by the confusion matrix, which classifies diabetic and non-diabetic people much better because of its equally good predictive capability. With an AUC of around 0.8102, this score is solidly strong for health survey data. The model also puts forth the importance of predictors in terms of its feature importance metric, thus acting in concert with expectations since it points to BMI, high blood pressure, and difficulty walking as some of the key contributors. Overall, Random Forest is robust, reliable, and well-suited for uncovering complex interactions across variables.

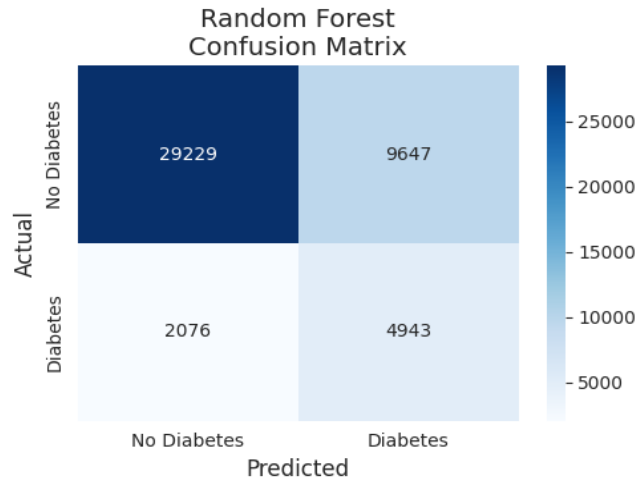

Fig. 15. Confusion matrix of Random Forest model.

D. GRADIENT BOOSTING

Gradient Boosting turned in the best performance in this study. It constructs trees one after another, each new tree trying to correct the errors of the previously built trees. This sequential learning allows the model to learn subtle patterns and intricate relationships that may not be captured by simpler models. Evaluation showed Gradient Boosting with an accuracy of about 85% and AUC of 0.8156, ranking as the very best among all metrics. The confusion matrix shows the highest ability of correctly identifying diabetics and the lowest rates of misclassifying them. The feature importance analysis showed that the strongest predictors are BMI, high blood pressure, high cholesterol, difficulty walking, and age category. Gradient Boosting combines high predictive power with interpretability via these importance measures, becoming the most effective method for this dataset.

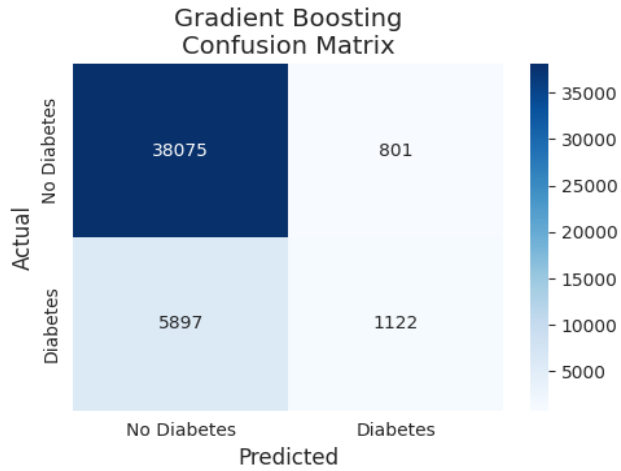

Fig. 16. Confusion matrix of Gradient Boosting model.

E. ROC Curve Comparison And Summary Metrics

The ROC curve analysis provided a thorough evaluation of the capabilities of all the models to classify both diabetic and non-diabetic subjects. The plots showed that the Gradient Boosting model performed the best. Its plot always remained above the plot of the rest of the models. This clearly denotes a robust ranking capability in distinguishing between the two classes. Random Foresting and Logistic Regression performed well compared to the rest. Their plots always remained above the threshold for almost all values of the classification threshold. They also had AUC values slightly over 0.80. The model that performed the worst was the Decision-Tree model. Its plot remained below the diagonal reference plot. This denotes a performance above random guessing. These plots validate the evaluation that ensemble concepts perform

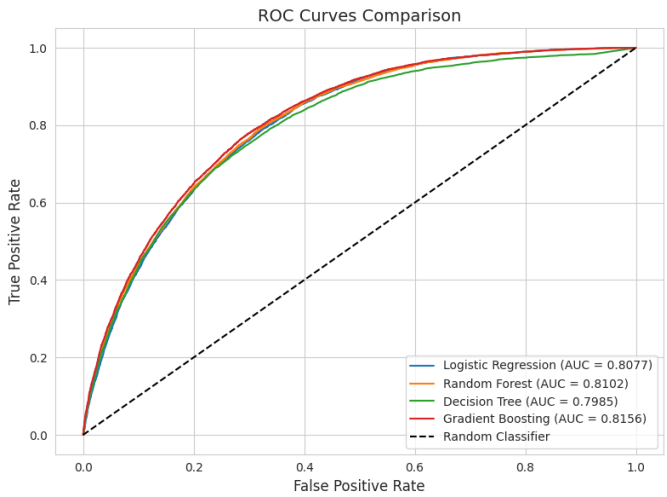

Fig 17. ROC curve comparison across all models.

The summary of accuracy and AUC values further helped in providing evidence for the results obtained in the ROC curve analysis. Gradient Boosting resulted in the highest accuracy and AUC value of 0.8156. Random Forest and Logistic Regression came close to the top three. The lowest accuracy and AUC values resulted in the Decision Tree classifier, further confirming that a single tree classifier lacks the generalization capability required for a particular dataset. These test results further validate the efficiency of ensemble techniques for modeling interactions of health indicators.

Model	Accuracy	AUC
Gradient Boosting	0.854058	0.815628
Random Forest	0.744569	0.810225
Logistic Regression	0.709947	0.807690
Decision Tree	0.684432	0.798484

Fig 18. Comparison of model accuracy and AUC scores.

The feature importance results from random forest added interpretability to the analysis by identifying which health indicators contributed most to predicting diabetes. BMI high blood pressure high cholesterol difficulty walking and age category emerged as the strongest predictors. This ranking aligns closely with established clinical literature and provides meaningful insight into the underlying health factors associated with diabetes. Including feature importance not only enhances model transparency but also strengthens the overall validity of the study conclusions.

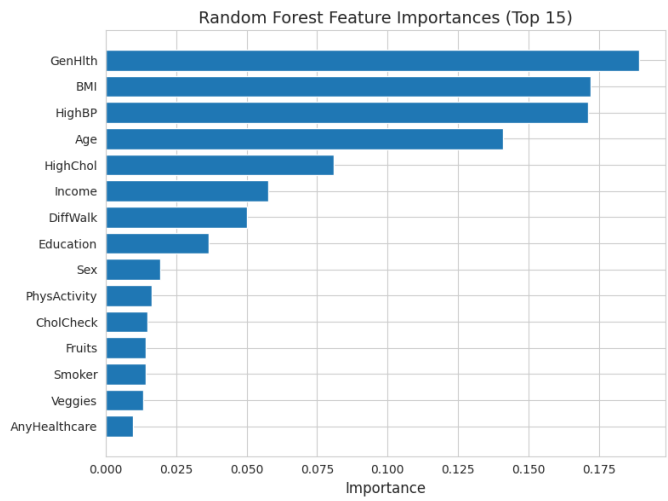

Fig 19. Feature importance values from the Random Forest model.

V. RESULTS AND DISCUSSION

The results of the analysis provide strong evidence that diabetes risk can be effectively predicted using self-reported health indicators from large scale surveys. The ensemble models in particular Gradient Boosting demonstrated high predictive accuracy and strong discriminatory ability. The results indicate that machine learning can identify meaningful patterns in non-clinical health data and classify individuals with a high degree of reliability. The behavioural and medical

indicators identified as significant contributors such as BMI high blood pressure high cholesterol and mobility limitations align closely with established clinical literature. This consistency supports the validity of the findings.

The study also shows that demographic factors contribute to diabetes risk but are not the primary drivers. Age plays a significant role since diabetes becomes more common with advancing age. Education and income showed indirect associations that explain certain lifestyle patterns but were not among the strongest predictors compared with direct medical indicators. This suggests that machine learning models benefit most from features representing physiological and functional health.

The ROC curves and confusion matrices provide complementary information. The ensemble models excel at identifying both negative and positive cases while Logistic Regression offers a strong baseline with interpretable coefficients. The Decision Tree performed the weakest although it still captured some meaningful patterns. Overall the models confirmed that diabetes prediction is feasible with self-reported survey data and that more complex models provide better accuracy and discrimination.

VII. CONCLUSION

The purpose of this study was to explore whether machine learning models can accurately predict diabetes risk using health lifestyle and demographic indicators from the CDC Diabetes Health Indicators dataset. The findings show that machine learning methods particularly Gradient Boosting can achieve strong predictive performance even when relying solely on self reported survey data. The study identified key predictors including BMI high blood pressure high cholesterol difficulty walking and age category which align with well documented clinical research and validate the integrity of the dataset.

This research demonstrates both descriptive and predictive value. The descriptive component reveals how health behaviors and demographic variables relate to diabetes. The predictive component shows how machine learning can use this information to classify individuals with high accuracy. While the dataset provides extensive coverage it also has limitations including reliance on self reported information which may introduce measurement errors. The absence of clinical measures such as blood glucose values limits the precision of medical interpretation.

Future work could incorporate advanced ensemble techniques such as XGBoost explore deep learning approaches analyze multiyear BRFSS data or integrate survey and clinical datasets to create hybrid predictive systems. Despite limitations the study shows that large scale self-reported health surveys can effectively support early detection strategies for diabetes and contribute to population health monitoring.

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